

Lecture Notes 4

TFP and Long-Term Economic Growth

Intermediate Macroeconomics, EC2211

Martin Flodén

Fall 2025

Department of Economics, Stockholm University

Contents and literature

- Towards endogenous growth
- The Romer model
- Institutions and growth
- Growth at the frontier

Literature:

- Jones (2024), chapter 6
- The Royal Swedish Academy of Sciences (2018), "[Integrating nature and knowledge into economics](#)", in particular pages 1-4
- Royal Swedish Academy of Sciences (2024) "[They provided an explanation for why some countries are rich and others poor](#)"
- Gordon (2014), "[The demise of U.S. economic growth](#)", sections 1-3.3
- Jones (2023), "[The outlook for long-term economic growth](#)"
- Draghi (2024) "[The future of European competitiveness](#)", pages 5-19

Towards endogenous growth

Towards endogenous growth

- The Solow model offers valuable insights into cross-country differences in income per capita but fails to explain sustained growth
- Next: endogenous growth

The AK "model"

What happens when we relax the assumption of diminishing marginal product of capital?

$$Y_t = AK_t \quad (1)$$

Capital accumulation:

$$K_{t+1} = (1 - \delta)K_t + I_t = (1 - \delta)K_t + sAK_t \quad (2)$$

Equations (1) and (2) imply that:

$$\frac{Y_{t+1}}{Y_t} = \frac{K_{t+1}}{K_t} = 1 - \delta + sA \quad (3)$$

The AK "model"

Since

$$\gamma \equiv \frac{Y_{t+1} - Y_t}{Y_t} = \frac{Y_{t+1}}{Y_t} - 1,$$

the growth rate of GDP is:

$$\gamma = sA - \delta \tag{4}$$

An assumption of constant marginal product of capital thus generates endogenous growth!

Recall that a higher savings rate s did *not* generate higher long-run growth in the Solow model

We see here that higher savings (more investments) would generate higher long-run growth if the marginal product of capital were not diminishing

The Romer model

The Romer model

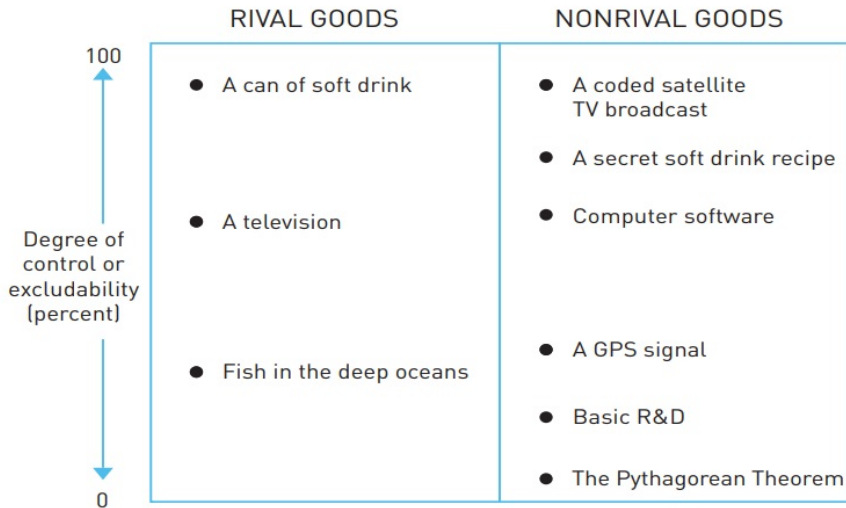
- Romer (1990), "Endogenous technological change"
- Nobel prize 2018: "for integrating technological innovations into long-run macroeconomic analysis"



Paul Romer 1955-

The Romer model

- Distinguish between objects (rivalrous) and ideas (non-rivalrous)
- Workers can either produce consumption goods or new ideas
- A single idea can be used by many people simultaneously
- Non-rivalry generates increasing returns
- Omit physical capital for simplicity



From Royal Academy of Sciences (2018)

Notation

Y_t : output of the consumption good

A_t : the existing stock of knowledge

L_t : the population

n : population growth rate

L_{yt} : number of workers employed in goods production

L_{at} : number of researchers (producing ideas)

γ : parameter in the production function

λ : fraction of labor force employed in research

ζ : productivity parameter

A_0 : the initial stock of knowledge

The Romer model

Production of the consumption good:

$$Y_t = A_t^\gamma L_{yt} \quad (5)$$

Production of new ideas:

$$\Delta A_{t+1} = \zeta L_{at} \quad (6)$$

Resource constraint for labor:

$$L_{yt} + L_{at} = L_t \quad (7)$$

and $L_{t+1} = (1 + n)L_t$

Assume that labor is allocated according to:

$$L_{at} = \lambda L \quad (8)$$

$$L_{yt} = (1 - \lambda)L \quad (9)$$

Solving the model

Dividing (5) by L and using (9) gives output per capita:

$$y_t \equiv \frac{Y_t}{L_t} = A_t^\gamma (1 - \lambda) \quad (10)$$

Dividing (6) by A_t gives the growth rate of TFP:

$$g_{At} \equiv \frac{\Delta A_{t+1}}{A_t} = \zeta \frac{L_{at}}{A_t}$$

or (assuming that the growth rate is positive)

$$A_t = \frac{\zeta}{g_{At}} L_{at} \quad (11)$$

The balanced growth path

- Is there a long-run equilibrium ("balanced growth path") where all variables grow at constant rates?
- If so, g_{At} must be constant (since we assume that L , L_y and L_a grow at constant rates)

Equation (11) then shows that the stock of knowledge is proportional to the number of researchers, L_{at}

- Since L_{at} in turn is a constant fraction of the population, this means that A and L must grow at the same rate on the balanced growth path

That is, $g_A = n$ in the long run

Interpreting the Romer model

- Equation (10) shows that output per capita will grow as long as the stock of knowledge, A_t increases
- The model moreover shows that the stock of knowledge increases at the same rate as the population
- And from the production function we see that the growth rate of output per capita is γn
- Note the key difference to the Solow model where an investment generates more capital, but where capital is rivalry – it is "less useful" when shared by many workers
- In the Romer model, investment generates more knowledge which is not rivalry; all workers become more productive irrespective of how many they are

A permanent rise in research productivity (higher ζ)

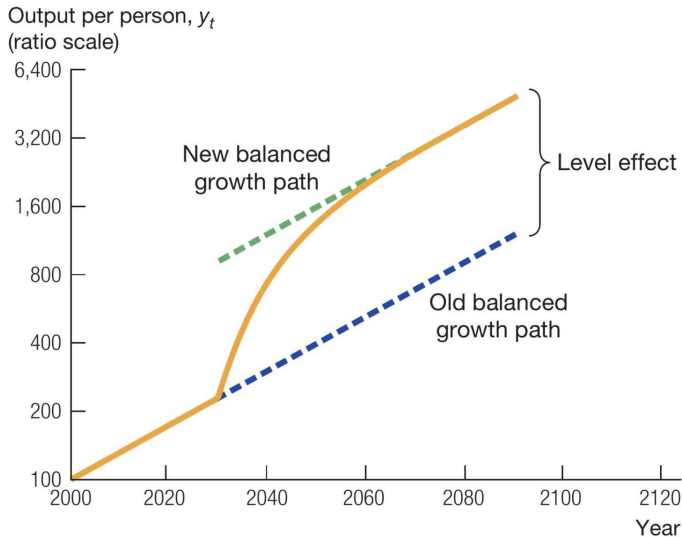


Figure 6.3 in Jones (2024). Recall that his $\bar{z}$ is our ζ .

A permanent rise in the research share (higher λ)

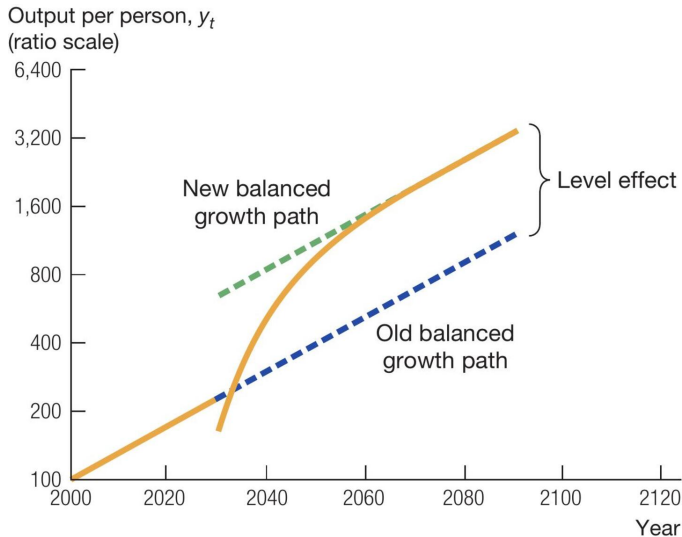


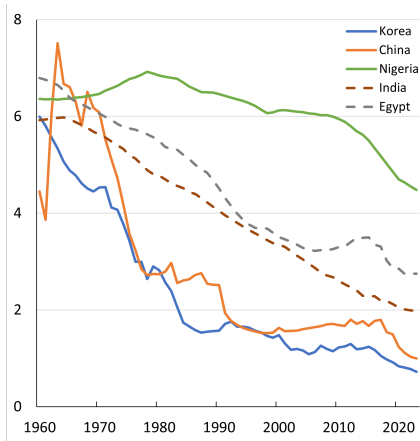
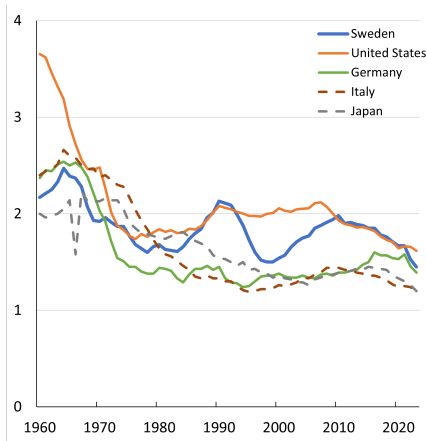
Figure 6.4 in Jones (2024). Recall that his $\bar{l}$ is our λ .

Summing up:

Malthus $\rightarrow$ Solow $\rightarrow$ Romer

- What is important in production?
Land (Malthus) → Physical capital (Solow) → Ideas (Romer)
- Population growth rates

Total fertility rates have trended down



Total fertility rates (the number of children that would be born to a woman if she were to live to the end of her childbearing years and bear children in accordance with age-specific fertility rates of the specified year). Source: [World Bank](#)

- What is important in production?
Land (Malthus) → Physical capital (Solow) → Ideas (Romer)
- Population growth rates
 - Malthus: Population growth is *endogenous*. (Do falling birth rates indicate that productivity has fallen?)
 - Solow: A lower population growth rate means that we do not have to build as much capital for our children
 - Romer: A lower population growth rate means that there will be fewer new ideas

Institutions and growth

The role of institutions

"The factors we have listed (innovation, economies of scale, education, capital accumulation, etc.) are not causes of growth; they are growth."

-North, D.C. and Thomas, R.P. (1973),
The Rise of the Western World: A New Economic History, Cambridge
University Press, Cambridge, UK.

Proximate versus fundamental causes of growth

- If growth is about factor accumulation and innovation, why do not poor countries invest and innovate?
- Why do countries provide different incentives to firms and households?
- Proximate causes of growth: factor accumulation and investment in R&D
- Fundamental causes of growth: luck? geography? culture? institutions?

Institutions and growth (Acemoglu, Johnson and Robinson)

- Institutions
 - Humanly devised set of rules
 - Place constraints on individual behaviour
- Economic institutions
 - Property rights
 - The presence and functioning of markets
 - Contractual opportunities available to firms and households
 - Entry barriers
- Political institutions
 - Polity and electoral laws
 - Checks and balances on political leaders

Democracy and income

- Is democracy conducive to high income?
- All the old democracies of the world are rich
- Growth performance of autocracies highly heterogeneous
 - Dictatorships excel or fall behind
 - Many of the East Asian growth miracles started out autocratic
- Complex relationship between polity and income!

Growth at the frontier

Economic growth: summing up

Sources of temporary and/or long-run growth:

- Physical capital
- Labor, hours worked
- Human capital
- Productivity, technology
 - Catching up with the frontier
 - Efficient use of resources and people
 - New ideas, research

Growth accounting: Can we quantify the contribution of different sources?

Consider our Cobb-Douglas production function:

$$Y_t = A_t K_t^\alpha L_t^{1-\alpha}$$

Apply the rules for growth rates:

$$g_{Y_t} = g_{A_t} + \alpha g_{K_t} + (1 - \alpha) g_{L_t}$$

In per capita terms, production is $y_t = A_t k_t^\alpha$, so

$$g_{y_t} = g_{A_t} + \alpha g_{k_t}$$

Growth accounting for the USA

TABLE 6.2

Growth Accounting for the United States

	1948–2021	1948–1973	1973–1995	1995–2003	2003–2021
Output per hour, Y/L	2.4	3.3	1.5	3.2	1.8
Contribution of K/L	0.9	1.0	0.8	1.4	0.8
Contribution of labor composition	0.2	0.2	0.2	0.3	0.3
Contribution of TFP, A	1.2	2.1	0.5	1.5	0.7

The table shows the average annual growth rate (in percent) for different variables.

Source: Bureau of Labor Statistics, *Multifactor Productivity Trends*.

Table 6.2 in Jones

Growth accounting with human capital

Suppose that the production function is:

$$Y_t = A_t K_t^\alpha (h_t L_t)^{1-\alpha}$$

where h is human capital per worker

In per capita terms:

$$y_t = A_t k_t^\alpha h_t^{1-\alpha}$$

Then:

$$g_{y_t} = g_{A_t} + \alpha g_{K_t} + (1 - \alpha) g_{h_t}$$

Growth accounting with human capital

Can we measure human capital or do we have to put it in the unexplained TFP term?

Human capital is often measured by the average number of years of schooling in the adult population. This is then converted to a human capital index based on empirical data on how schooling affects wages.

The [Penn World Tables](#) report such an index:

	Years of schooling	"Human capital"
USA	13.6	3.75
Sweden	12.3	3.44
Mexico	9.2	2.78
Malawi	5.7	0.71

The outlook for long-term economic growth

Growth in the U.S. has been surprisingly stable at 2% per year for the past 150 years

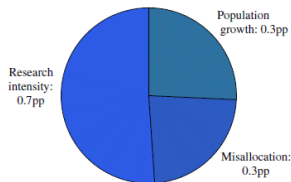
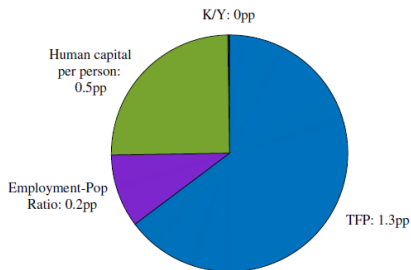
Gordon (2014) and Jones (2023) note that much of this growth is a result of

- Rising educational attainment
- Rising investment in research

I would add globalization (with offshoring, restructuring etc) to the list

The outlook for long-term economic growth

**Components of 2% Growth
in GDP per Person**



Components of 1.3% TFP Growth

Note: The figure shows a stylized growth accounting exercise for the United States since the 1950s.

Figure 5 in Jones (2023)

The outlook for long-term economic growth

Potential tailwinds:

- Better utilization of resources (use of talents)
- More frontier contribution from China and India
- AI

Headwinds:

- Educational attainment cannot/will not continue to increase
- Research intensity cannot/will not continue to increase
- Population growth will slow or turn negative
- Geopolitics, end of globalization

What we did (growth part of the course)

- Growth facts
- Growth accounting
- From Malthus to Solow to Romer
- Institutions